

RYAN JAMES F. INDANGAN

AI Software Engineer • Full-Stack Developer • Certified CTO

CURRICULUM VITAE

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UpWork • LinkedIn • GitHub Portfolio

PROFESSIONAL SUMMARY

AI Software Engineer, full-stack developer, and Certified CTO with 8+ years of experience spanning AI/ML engineering, end-to-end product development, and technical leadership. Builds **health-technology platforms** — clinical-trial intelligence, registry data engineering, and AI-assisted business development — and ships document-intelligence and LLM-powered systems — OCR, computer vision, ML-based extraction, watermark removal, and underwriting automation — alongside secure, scalable web platforms across the LAMP and modern JavaScript stacks. Has delivered for clinical research (Core Clinical Trials), financial services (Alternative Funding Group), government and United Nations initiatives, and 50+ international clients as a Top Rated freelancer. A startup co-founder who has led cross-functional teams of up to 12 engineers from architecture through deployment, pairing deep hands-on engineering with strategic, business-aligned thinking.

TECHNICAL SKILLS

Health Technology & Clinical Research

- Clinical-trial intelligence platforms: study/site/investigator entity modelling, registry ETL at scale (150k+ site records), entity resolution and deduplication, field-level provenance, and executive reporting
- Clinical-trial business development: pipeline tracking, opportunity scoring, registry-driven lead discovery
- ClinicalTrials.gov integration — live status synchronisation, phase-change detection, registry reconciliation
- Audit & integrity engineering: append-only SHA-256 hash-chained records with scheduled chain verification, role-based access, TOTP 2FA, reversible data merges
- AI research intelligence — cited, confidence-rated background briefs generated over trial and market data

AI & Machine Learning

- Document intelligence & OCR (Tesseract, OpenCV, PyMuPDF, pdfplumber, pdf2image); computer vision and image processing
- Machine learning (scikit-learn, Random Forests, feature engineering); ML-based watermark detection & removal
- LLM integration & orchestration (Ollama, OpenAI-compatible APIs, vLLM); conversational AI & companion applications
- End-to-end AI pipeline design, validation, and real-time streaming (FastAPI Server-Sent Events); model deployment (Hugging Face Spaces)

Web, API & Real-Time Systems

- Full-stack development: React, Next.js, Vue, Angular, Node.js, TypeScript, PHP, Python
- RESTful & GraphQL APIs, WebSockets, Server-Sent Events; real-time and payment/transaction-processing systems
- Secure, production-ready services: JWT/RSA auth, bcrypt, rate limiting, CORS, hardened API gateways

Workflow Automation & Integration

- Process & workflow automation (n8n); CI/CD automation; data transformation and ETL-style pipelines
- Legacy-to-custom migrations (e.g., Salesforce → bespoke web applications); REST and third-party integrations
- Spreadsheet/reporting automation (openpyxl); document generation and templating

Frameworks & Libraries

- FastAPI, Express.js, Laravel, CodeIgniter 3, Django
- OpenCV, scikit-image, Pillow, Tesseract OCR, pdfplumber, pdf2image, PyMuPDF, openpyxl, NumPy / pandas

Cloud, DevOps & Databases

- AWS (EC2, RDS, Lambda, EventBridge, S3, CloudFront), DigitalOcean; Docker, Docker Compose, Kubernetes
- CI/CD (GitHub Actions, Jenkins, Bitbucket Pipelines); MySQL, PostgreSQL, SQLite, MongoDB, Redis, Firebase

Programming, Testing & Collaboration

- Python, JavaScript/TypeScript, PHP, SQL, HTML/CSS, Bash; Git, GitHub/GitLab, Postman, Insomnia
- Pytest, Jest, Mocha, Selenium, Cypress; Jira, Confluence, Asana, Trello, Slack, MS Teams, ClickUp, Monday.com

PROFESSIONAL EXPERIENCE

HireOverseas (deployed to Core Clinical Trials) — AI Workflow Engineer

May 2026 – Present

Remote **Independent Contractor**

- Engaged through HireOverseas and deployed to Core Clinical Trials as lead developer across four systems — two internal clinical-trial intelligence platforms and two public-facing marketing sites — owning each end to end from requirements and architecture through build, deployment tooling, and release.
- **Clinical Trial Intelligence & BD Platform (internal name: CDMS)** — architected and shipped the organization's internal trial-intelligence and business-development platform on AWS (Elastic Beanstalk, RDS), replacing a Salesforce-based study tracker with a tailored web application that eliminates licensing overhead and matches the team's exact workflows.
 - Models trials, sites, investigators, and sponsors keyed on the ClinicalTrials.gov NCT ID, so registry trials, imported tracker rows, and sponsor contacts meet on one join — making "are we already pursuing this trial?" a single lookup rather than forcing the domain into a generic CRM object.
 - Tamper-evident audit trail with a hash-chained daily verification job that flags any altered record.
 - Role-based access control, two-factor authentication, and encrypted application settings.
 - Automated migrations on deploy, scheduled jobs and queue workers, daily snapshots with point-in-time recovery.
- **BD Pipeline** — the clinical-trial business-development engine within the CDMS, turning scattered spreadsheets, public registry data, and market signals into a single ranked pipeline of trials the organization can bid to run.
 - Study-tracker spreadsheet ingestion with defensive parsing and idempotent, null-skipping upserts that never clobber existing data.
 - Live ClinicalTrials.gov status synchronisation that flags when a tracked study's registry status moves, with an acknowledge-and-re-snapshot step so the same change is never surfaced twice.
 - Opportunity discovery from SEC filings and news signals alongside registry trials.
 - Polymorphic lead-tracking CRM — contact log, comments, and owner-aware follow-up reminders — shared across every lead surface.
 - AI-assisted outreach drafting with a cross-source duplicate-contact guard that surfaces every colliding study and prior touch on a shared inbox before anyone hits send.
- **Core Research Intelligence** — built a standalone site-intelligence platform with its own repo, composer package, ~27-table schema, and admin panel, mining the public trial registry for the research sites and investigators worth contacting.
 - Resumable, checkpointed ClinicalTrials.gov ETL that pulled 21,288 studies across 73 condition queries in a single 59-minute live run.
 - Entity resolution across studies, sites, investigators, and sponsors over a 150,000-site corpus, with raw payloads and field-level provenance retained.
 - Explainable rules-based site classification and a 0-100 opportunity score whose weights are retunable at runtime without a deploy.
 - Blocked-pair duplicate detection with human review and reversible merges — never auto-merging on a fuzzy name match alone; covered by 186 tests on a SQLite + MySQL CI matrix.
- **AI research enrichment** (within the CDMS) — a reversible AI background-research layer compiling cited context briefs on trials, sponsor companies, and market signals.
 - Every priority trial and market signal carries a generated background brief — web-researched with real cited sources, a self-assessed confidence rating, and explicit notes on what could not be verified.
 - Additive by design — enrichment never mutates the source record it describes.
 - Fully reversible: feature-flagged off, or cleared by batch, type, or individual record.
 - Surfaced dozens of trials whose registry status had silently gone stale as a side effect.
- **TrialPulse** — delivered a marketing and demo platform with a custom CMS engine featuring full edit history, one-click revert, and sandboxed preview, so non-technical staff can safely edit the live site themselves; includes CMS-editable demo-request forms that deliver enquiries straight to the team's inbox, with nothing stored on the site.
- **Corporate site rework** — rebuilt the organization's public website from the ground up on a new enterprise design system, fully responsive and staged behind a private preview so the live domain stayed untouched until vetted.

- Automating labor-intensive manual processes for the CEO and Executive Director, reducing operational workload and turnaround time.
- Designing AI-assisted ingestion and data pipelines — Claude-assisted spreadsheet column mapping that survives hand-reordered exports, backed by a deterministic fixed-column fallback — with every model call routed through a single cost-logged, audited service restricted to read-only data access.

FirstKind AI — AI Software Engineer & Co-Founder, GO Models

Jan 2026 – Present

Remote **Independent Contractor**

- Co-founded GO Models, a suite of AI companion applications, serving as founding AI Software Engineer.
- Architect and build LLM-powered conversational experiences end-to-end — model integration, backend APIs, and frontend.
- Drive product and technical direction from prototype to deployment, establishing engineering standards and AI infrastructure.

Alliance Global Solutions BPO Intl Corp. — AI Developer / Machine Learning Engineer

Nov 2025 – Apr 2026

Alternative Funding Group · On-site — SM North EDSA, Tower 1, Quezon City **Full-time**

- Built an end-to-end bank statement extraction and validation pipeline combining native PDF text extraction with OCR fallback to support automated underwriting workflows.
- Implemented advanced OCR preprocessing using OpenCV and PIL — including denoising, adaptive thresholding, and deskewing — with aggressive multi-angle OCR for difficult scans selected via quality scoring.
- Designed a watermark-removal ML subsystem using feature engineering and RandomForest classification to auto-detect watermark types and route documents to appropriate removal strategies.
- Created a validation-first extraction framework enforcing structural integrity invariants and auditability without mutating source financial text.
- Integrated local LLM workflows via Ollama with OpenAI-compatible backends to generate structured underwriting summaries; built a secure Node gateway with JWT/RSA management and FastAPI Server-Sent Events for real-time processing visibility.

Upwork — Full-Stack Developer (Top Rated)

Apr 2021 – Oct 2025

Remote **Freelance Platform**

- Achieved Top Rated status with a 100% Job Success Score across 50+ delivered projects for international clients.
- Specialized in the LAMP stack and modern JavaScript frameworks, maintaining long-term partnerships with recurring clients.
- Delivered several of the contract engagements listed in this CV through the Upwork platform.

Bada LLC — Support Engineer

Apr 2025 – Oct 2025

Remote

- Developed and scripted solutions for data management and client image migrations, improving operational efficiency.
- Created custom templates for InDesign, Word, and PowerPoint; integrated systems using REST APIs and JavaScript for cross-platform automation.
- Troubleshoot server-side, end-user, and codebase issues and managed client support tickets from initiation to resolution.

Skills: JavaScript (ES5), REST APIs, ZenDesk, InDesign, Windows Server, Linux Server, JSON

GlobalX Digital Corporation — Consultant

Jan 2025 – Apr 2025

Hybrid — Pasay City, NCR

- Led the redesign of database architecture, creating scalable, high-performing systems that improved data integrity and business alignment.
- Developed and secured API endpoints; directed 12 developers across 3 teams, providing technical guidance, architectural leadership, and workflow documentation.
- Revived a stalled project by assuming ownership, delivering a fully enhanced Version 2 within 2 months after 9 months of failed prior leadership.
- Acted as Technical Lead, aligning goals among cross-functional teams and stakeholders to ensure delivery within strict timelines.

Net Global Solutions Inc. — Back End Developer / Project Lead

Apr 2023 – Nov 2024

Hybrid — Pasig, Metro Manila

- Served as Project Lead & Lead Programmer for the PAGCOR compliance system (Back Office Access), ensuring regulatory adherence.
- Managed the PEZA Digital Marketplace (Builder.ai) as Project Manager, enabling streamlined data management for the Philippine Economic Zone Authority.
- Built a custom PHP framework to enhance API development and system performance.
- Developed a Sports Data Management System, an Automated Billing System, and multiple enterprise solutions as Lead Programmer.

Intermobile Solutions Corporation — Chief Technology Officer / Co-Founder

Sep 2023 – Jan 2024

Hybrid — Makati City, Metro Manila

- Spearheaded product design and technical direction for early-stage startup projects.
- Developed proof-of-concept applications and internal systems to demonstrate product viability to investors.
- Established foundational infrastructure, workflows, and code standards for rapid prototyping and small-scale deployment.
- Provided technical mentoring and handled initial backend and API integrations for pilot products.

Microbitz Solutions Inc. — IT Director / Full-Stack Software Engineer

Jul 2023 – Jan 2024

Hybrid — Makati City, Metro Manila

- Led full-cycle development of an online lottery platform, from architecture planning to deployment and maintenance.
- Designed and managed system infrastructure, ensuring high availability, scalability, and data security.
- Developed and optimized backend APIs for secure transaction processing; built and maintained the database schema with reliable integrity and backup strategies.
- Oversaw technical direction, guided developers, and implemented CI/CD processes and server monitoring to maintain uptime.

ADEC Innovations — Systems Developer II

Mar 2022 – Apr 2023

Remote

- Developed the UN-Habitat SDG Cities platform as part of a United Nations initiative.
- Built the Disaster Risk Reduction Management Information System for the Department of Education (DepEd).
- Created a database automated-remapping system ("Bucket Service") to optimize DepEd's database management.

Freelance — Web Developer

2019 – 2023

Remote

- Designed and developed a Sports Data Management System for a private company.
- Built an Online Human Resource Information System for STI Muñoz EDSA.
- Delivered custom websites and applications for small businesses and organizations.

Remote

- Directed large-scale web and mobile projects including a Lotto Play Mobile App, Wholesale E-Commerce, and white-labeled Learning Management Systems.
- Built enterprise systems: Online Inventory & Asset Tracking, Package Tracking, Broker Accreditation (Ayala Land), and Log Monitoring with Digital Signature.
- Delivered custom after-sales service and tracking solutions for corporate clients (Esquire Tech, Chubb Limited, Marmo Pizzara Trading).
- Provided technical leadership as Lead Programmer, ensuring quality delivery and scalable architecture across projects.

TEACHING, LEADERSHIP & EARLY EXPERIENCE

Sunrise Christian Academy — Vice Principal & Summer Mathematics Instructor (Elementary)

2016 – 2020

Family-owned school

- Served as Vice Principal, supporting academic coordination and day-to-day school operations.
- Taught Mathematics to elementary students during summer classes.
- Held the role alongside undergraduate Information Technology studies and early software development work.

Golden Sun Academy — Janitorial & Maintenance Staff

2010 – 2011

Family-owned school

- Maintained the cleanliness and orderliness of classrooms and facilities while in first year of high school.
- Built an early foundation of work ethic, reliability, and responsibility within the family-run institution.

EDUCATION

Bachelor of Science in Information Technology

STI College — Muñoz EDSA

- Relevant Coursework: Data Structures, Algorithms, Database Development, Programming.

Secondary Education

Graceland Christian College

- Graduated as CAT Corps Commander, leading and coordinating unit activities and drills.
- Member, Performing Arts Guild of Graceland.

SELECTED PROJECTS

Health Tech

- **CDMS — Clinical Trial Intelligence & BD Platform:** AWS-hosted trial-intelligence and BD platform keyed on ClinicalTrials.gov NCT IDs, with a SHA-256 hash-chained tamper-evident audit trail, RBAC and TOTP 2FA.
- **Core Research Intelligence:** Site-intelligence platform with a resumable ClinicalTrials.gov ETL (21,288 studies in 59 min), entity resolution over 150k sites, explainable scoring, and 186 tests on a SQLite+MySQL CI matrix.
- **BD Pipeline & TrialPulse:** A BD engine unifying spreadsheets, registry data and SEC/news signals into a ranked pipeline with a polymorphic lead CRM and AI-assisted outreach; plus a marketing site on a custom CMS engine with edit history and one-click revert.

AI & Machine Learning

- **GO Models:** Co-founded and engineered a suite of LLM-powered AI companion apps end to end as founding AI Software Engineer.
- **Watermark-Removal ML Subsystem:** RandomForest/feature-engineering subsystem that auto-detects watermark types and routes documents to the right removal strategy.

Fin Tech

- **Bank Statement Extraction & Underwriting Pipeline:** End-to-end extraction/underwriting pipeline — native PDF + OCR fallback, OpenCV preprocessing, deterministic classification, local-LLM summaries via Ollama, FastAPI SSE + Node/JWT gateway, React UI.
- **Payment Gateway & Crypto Checkout:** A Django/PostgreSQL payment-gateway backend (merchants, payments, transaction tracking; client-confidential) and an interactive end-to-end crypto checkout flow.

Gaming

- **ARMADO:** Original browser tactical FPS (three.js client + Node/WebSocket server) with a Filament back office featuring a 3D map builder, live weapon tuning and content studios — 100% original code, procedural art and audio.
- **LAAS — Procedural WebGPU World Engine:** Fully procedural in-browser open-world engine (WebGPU) generating every mesh, texture, light and sound from code, with zero bundled assets.
- **iGaming & Betting Platforms:** Multiple real-money gaming backends — casino/lottery, core gaming API, horse-racing and live sabong — with operator back-offices, wallets and crypto payments (client-confidential).

Gov Tech

- **PAGCOR Compliance System:** Led a Symfony/Doctrine regulatory-compliance system for the Philippine gaming regulator PAGCOR (client-confidential).
- **UN-Habitat SDG Cities · DepEd DRRMIS · PEZA Marketplace:** The UN-Habitat SDG Cities platform (United Nations), the Disaster Risk Reduction MIS for the Department of Education, and the PEZA Digital Marketplace (Builder.ai) as Project Manager.

Enterprise, Web & Dev Tools

- **Kayanin & Faith Temple:** A Laravel 13 Philippines-first freelance marketplace with skills-exam qualification, and a Laravel 13 corporate CMS + church-management platform with a fully editable public site.
- **GlobalX, Broker Accreditation (Ayala Land) & more:** Enterprise platforms across real estate, HR (HRIS) and education (LMS).
- **LastResortAI:** A multi-agent dev crew (plan → build → adversarial review → deploy) with persistent memory and multi-pass verification on Claude, Claude Code and the Claude Agent SDK; plus a custom PHP framework and CI/CD automations.

AWARDS & ACHIEVEMENTS

- Top Rated Freelancer on Upwork — 100% Job Success Score across 50+ delivered projects.
- Led 12 developers across 3 teams at GlobalX Digital Corporation and revived a stalled project to deliver Version 2 in 2 months.
- Union Bank UHack 2017 — Semi-Finalist (December 2, 2017).
- HealthHacks PH 2017 — Semi-Finalist (February 17, 2017).
- Battle of the Debuggers 2015 — 1st Runner-up, College (October 2, 2015).

CERTIFICATIONS

Leadership & Management

- Certified Chief Technology Officer, Information Technology — Institute of Management, Technology and Finance (IMTF), 2023.

Artificial Intelligence (Anthropic)

- Claude Code in Action — Anthropic (2026). Verify: verify.skilljar.com/c/usugbinq3tyu
- Claude 101 — Anthropic (2026). Verify: verify.skilljar.com/c/ggyq4opwistp

Cybersecurity

- Ethical Hacking Course (Hacker-X) — 21-module program (Sep 2024 – Nov 2025): Know the OS; Hack the Mac; Gathering Information; Leveraging AI for Hacking; Prompt Guide for Hacking; Surfing Anonymously; Hiding Messages; Social Media Hacking; Credit/Debit Card Frauds; Keyboard Spying; WiFi Hacking (WEP, WPA/WPA2); Network Spying; Database Hacking; Android Hacking; Bringing Down a Website; Cross-Site Scripting; Make Your Data Safe; Hacking Passwords; WordPress Scanning; and Vulnerability Scanning & Reporting.

Programming Fundamentals (SoloLearn)

- Python Core · Java · JavaScript · C# · PHP · SQL · HTML · CSS — 8 SoloLearn course certificates (2018–2023), each verifiable at sololearn.com/certificates.

REFERENCES

Available upon request.